ASSIGNMENT-6

Task – 1: Networking Fundamentals

SOLUTION-

Networking is the process of connecting two or more computers or devices so that they can communicate and share information. In Linux, networking is managed using different interfaces, IP addresses, routing tables, and configuration files. Linux provides several commands such as ip addr, ip route, and nmcli device show to display and manage network information.

1. IP Address (Internet Protocol Address)

Definition

An **IP (Internet Protocol) Address** is a unique numerical address assigned to every device connected to a network. It acts like the postal address of a computer, allowing devices to send and receive data over a network.

Without an IP address, a computer cannot communicate with other devices on a local network or the Internet.

There are two versions of IP addresses used in Linux:

- IPv4
- IPv6

➤ IPv4 (Internet Protocol Version 4)

IPv4 is the most widely used version of the Internet Protocol.

Features

- Uses **32-bit** addressing.
- Written as **four decimal numbers separated by dots**.
- Each number ranges from **0 to 255**.
- Supports about **4.3 billion unique addresses**.

Example

172.20.206.173

➤**IPv6 (Internet Protocol Version 6)**

IPv6 was introduced to overcome the address shortage of IPv4.

Features

- Uses **128-bit** addressing.
- Written in hexadecimal format separated by colons (:).
- Provides an extremely large number of addresses.

Example

fe80::215:5dff:fe55:4d11/64

2. Subnet Mask

Definition

A **Subnet Mask** is a 32-bit number used with an IPv4 address to divide a network into smaller networks called **subnets**.

It helps the operating system determine:

- Which part of the IP address identifies the **network**
- Which part identifies the **host (computer)**

Without a subnet mask, computers cannot determine whether another device is on the same network or a different network.

Example-

Equivalent Subnet Mask

255.255.240.0

3. Default Gateway

Definition

A **Default Gateway** is the router or network device that connects a local network to other networks, such as the Internet.

Whenever a computer wants to send data outside its own network, it forwards the data to the default gateway.

The gateway then sends the data to the correct destination.

Without a default gateway, a computer can communicate only with devices on the same local network.

Example

Default Gateway
172.20.192.1

4. MAC Address (Media Access Control Address)

Definition

A **MAC Address** is the permanent physical address assigned to a Network Interface Card (NIC) by the manufacturer.

Unlike an IP address, the MAC address usually does not change.

It is used for communication within the same local network.

Features

- 48-bit address.
- Written in hexadecimal.
- Every network card has a unique MAC address.

Example

00:15:5d:55:4d:11

5. Network Interface

Definition

A **Network Interface** is the hardware or software component that connects a computer to a network.

Linux uses different names for network interfaces.

Examples

- eth0 (Ethernet)
- wlan0 (Wireless)
- lo (Loopback)

Every interface has:

- IP Address
- MAC Address
- Network Status
- Example
- ```
  GENERAL.DEVICE: eth0
  GENERAL.TYPE: ethernet
  ```

➤Linux Networking Commands

•ip addr

Purpose

Displays detailed information about all network interfaces.

•nmcli device show

Purpose

Displays detailed information about network devices managed by NetworkManager.

• ip route

Purpose

Displays the routing table.

Task – 2: DHCP vs Static IP Configuration

SOLUTION-

In Linux, every computer connected to a network requires an **IP address** for communication. There are two common methods of assigning an IP address:

1. **DHCP (Dynamic Host Configuration Protocol)**
2. **Static IP Address**

➤DHCP-

DHCP (Dynamic Host Configuration Protocol) is a network protocol that automatically assigns IP addresses and other network settings to devices connected to a network.
  ```

Instead of configuring every computer manually, a **DHCP Server** automatically provides:

- IP Address
- Subnet Mask
- Default Gateway
- DNS Server

This makes network administration much easier.

Advantages of DHCP

- Automatic IP assignment
- Reduces manual configuration
- Prevents IP conflicts
- Easy to manage large networks
- Saves administrator time

Disadvantages of DHCP

- Depends on a DHCP server
- IP address may change
- Not suitable for servers requiring fixed addresses

➥DHCP Working Process (DORA)

DHCP works using a **4-step process** called **DORA**.

1. Discover (D)

When a computer joins the network, it does not have an IP address.

The client broadcasts a **DHCP Discover** message asking:

"Is there any DHCP server available?"

2. Offer (O)

The DHCP server receives the request and replies with a **DHCP Offer** message.

The offer contains:

- IP Address
- Subnet Mask
- Gateway
- DNS Server
- Lease Time

3. Request (R)

The client accepts the offered IP address and sends a **DHCP Request** message to the server.

This informs the server that it wants to use the offered IP address.

4. Acknowledge (A)

The DHCP server confirms the request by sending a **DHCP Acknowledge (ACK)** message.

The client now receives:

- IP Address
- Subnet Mask
- Gateway
- DNS Server

The computer is now ready to communicate on the network.

➤Static IP Address-

A **Static IP Address** is an IP address that is manually assigned by the system administrator.

Unlike DHCP, it does not change automatically after rebooting or reconnecting to the network.

Static IPs are commonly used for:

- Web Servers

- Database Servers
- DNS Servers
- Mail Servers
- Printers
- CCTV Systems

Advantages of Static IP

- Permanent IP address
- Reliable for servers
- Easy remote access
- Better for hosting websites and services
- Simplifies DNS and port forwarding

Disadvantages of Static IP

- Manual configuration required
- Time-consuming for many devices
- Incorrect settings may cause IP conflicts
- Requires careful network management

➥Configure Static IP Using nmcli (Ubuntu)-

Step 1: Check Network Interface

- **nmcli device status**

```
praveen@praveenasus:~$ nmcli device status
DEVICE   TYPE       STATE                   CONNECTION
eth0     ethernet   connected (externally)  eth0
lo       loopback   unmanaged               --
praveen@praveenasus:~$
```

Step 2: Show Current Connection

•nmcli connection show

```
praveen@praveenasus:~$ nmcli connection show
NAME                UUID                                  TYPE      DEVICE
eth0                76fcec3c-32af-4ed1-8727-bc913a49f038  ethernet  eth0
Wired connection 1  eba34725-4898-36eb-b146-026ee1391570  ethernet  --
praveen@praveenasus:~$
```

Step 3: Configure Static IP

Example:

•sudo nmcli connection modify "Wired connection 1" \
ipv4.addresses 192.168.1.100/24 \
ipv4.gateway 192.168.1.1 \
ipv4.dns 8.8.8.8 \
ipv4.method manual

Step 4: Restart Connection

•sudo nmcli connection down "Wired connection 1"

•sudo nmcli connection up "Wired connection 1"

Step 5: Verify IP Address

•ip addr

Example Output

inet 192.168.1.100/24

Task – 3: DNS Configuration & Testing

SOLUTION-

Introduction

DNS (Domain Name System) is a distributed naming system that translates human-readable domain names (such as **google.com**) into IP addresses (such as **142.250.193.14**). Since computers communicate using IP addresses, DNS acts like the **phonebook of the Internet**, allowing users to access websites using easy-to-remember names instead of numerical IP addresses.

Without DNS, users would have to remember the IP address of every website they want to visit.

➤DNS-

DNS (Domain Name System) is a network service that converts a domain name into its corresponding IP address. It allows computers and users to communicate easily over the Internet.

Example

Instead of typing:

142.250.193.14

we simply type:

google.com

DNS automatically converts the domain name into the correct IP address.

➥How Name Resolution Works

When you type **google.com** in your browser, Linux follows these steps:

1. The browser requests the IP address for google.com.
2. Linux first checks the **/etc/hosts** file for a matching entry.
3. If not found, Linux checks the DNS servers listed in **/etc/resolv.conf**.

4. The DNS server searches its records.
5. The DNS server returns the IP address.
6. The browser connects to the website using that IP address.

➤output of following commands in ubuntu-

1. View Current DNS Servers

Command

cat /etc/resolv.conf

```
praveen@praveenasus:~$ cat /etc/resolv.conf
# This file was automatically generated by WSL. To stop automatic generation of this
file, add the following entry to /etc/wsl.conf:
# [network]
# generateResolvConf = false
nameserver 10.255.255.254
praveen@praveenasus:~$
```

2. Resolve a Domain Using nslookup

Command = nslookup google.com

```
praveen@praveenasus:~$ nslookup google.com
Server:         10.255.255.254
Address:        10.255.255.254#53

Non-authoritative answer:
Name:   google.com
Address: 192.178.174.113
Name:   google.com
Address: 192.178.174.139
Name:   google.com
Address: 192.178.174.102
Name:   google.com
Address: 192.178.174.101
Name:   google.com
Address: 192.178.174.100
Name:   google.com
Address: 192.178.174.138
Name:   google.com
Address: 2404:6800:4002:81d::200e
```

3. Resolve a Domain Using dig

Command

dig google.com

```
praveen@praveenasus:~$ dig google.com

; <<>> DiG 9.20.18-1ubuntu2.1-Ubuntu <<>> google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 34831
;; flags: qr rd ra; QUERY: 1, ANSWER: 6, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4096
;; QUESTION SECTION:
;google.com.                    IN      A

;; ANSWER SECTION:
google.com.             100     IN      A       192.178.174.113
google.com.             100     IN      A       192.178.174.139
google.com.             100     IN      A       192.178.174.102
google.com.             100     IN      A       192.178.174.101
google.com.             100     IN      A       192.178.174.100
google.com.             100     IN      A       192.178.174.138

;; Query time: 19 msec
;; SERVER: 10.255.255.254#53(10.255.255.254) (UDP)
;; WHEN: Fri Jun 19 07:25:49 UTC 2026
;; MSG SIZE  rcvd: 135
```

4. Add a Custom Entry in /etc/hosts

Open the hosts file:

sudo nano /etc/hosts

```
praveen@praveenasus:~$ sudo nano /etc/hosts
```

```
praveen@praveenasus: ~
 GNU nano 8.7.1                          /etc/hosts
# This file was automatically generated by WSL. To stop automatic generation
# of this file, add the following entry to /etc/wsl.conf: [network]
# generateHosts = false
127.0.0.1       localhost
127.0.1.1       praveenasus.localdomain praveenasus

# The following lines are desirable for IPv6 capable hosts
::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
192.168.1.10    myserver.local
```

5. Order of Name Resolution

Linux decides where to search for names using the file:

sudo cat /etc/nsswitch.conf

```
praveen@praveenasus:~$ sudo cat /etc/nsswitch.conf
# /etc/nsswitch.conf
#
# Example configuration of GNU Name Service Switch functionality.
# If you have the `glibc-doc-reference' and `info' packages installed, try:
# `info libc "Name Service Switch"' for information about this file.

passwd:         files systemd
group:          files systemd
shadow:         files systemd
gshadow:        files systemd

hosts:          files dns
networks:       files

protocols:      db files
services:       db files
ethers:         db files
rpc:            db files

netgroup:       nis
praveen@praveenasus:~$
```

Meaning

files

Search /etc/hosts first.

If the hostname exists here, Linux immediately returns the IP address.

Task – 4: SSH Setup and Remote Access-

SOLUTION-

SSH (Secure Shell) is a secure network protocol used to remotely access and manage Linux systems over a network. It encrypts all communication between the client and the server, making it much safer than older protocols like Telnet. System administrators and DevOps engineers commonly use SSH to manage servers, transfer files, and execute remote commands.

Step 1: Install and Start the SSH Service

Update the package list

•sudo apt update

Install OpenSSH Server

•sudo apt install openssh-server

Check the SSH Service Status

•sudo systemctl status ssh

Start the SSH Service (if not running)

•sudo systemctl start ssh

Enable SSH at Boot

•sudo systemctl enable ssh

```
praveen@praveenasus:~$ sudo apt update
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Hit:2 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [31.1
 kB]
Get:5 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [
43.1 kB]
Hit:6 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:7 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [257 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [49.4 k
B]
Get:9 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [150
kB]
Get:10 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [6
2.2 kB]
Fetched 868 kB in 2s (482 kB/s)
14 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

```
praveen@praveenasus:~$ sudo apt install openssh-server
openssh-server is already the newest version (1:10.2p1-2ubuntu3.2).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 14
```

```
praveen@praveenasus:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
     Active: active (running) since Fri 2026-06-19 06:40:21 UTC; 23h ago
 Invocation: bce5793d7f5644cea688ac0924ca4372
TriggeredBy: ● ssh.socket
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 219 (sshd)
      Tasks: 1 (limit: 4333)
     Memory: 1.6M (peak: 2.4M)
        CPU: 161ms
     CGroup: /system.slice/ssh.service
             └─219 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jun 19 06:40:21 praveenasus systemd[1]: Starting ssh.service - OpenBSD Secure Shell >
Jun 19 06:40:21 praveenasus sshd[219]: Server listening on 0.0.0.0 port 22.
Jun 19 06:40:21 praveenasus sshd[219]: Server listening on :: port 22.
Jun 19 06:40:21 praveenasus systemd[1]: Started ssh.service - OpenBSD Secure Shell s>
lines 1-18/18 (END)
```

```
praveen@praveenasus:~$ sudo systemctl start ssh
praveen@praveenasus:~$ sudo systemctl enable ssh
Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/sys
temd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
praveen@praveenasus:~$
```

Step 2: Find the IP Address of the Linux VM

Command

ip addr

```
praveen@praveenasus:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qle
n 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet 10.255.255.254/32 brd 10.255.255.254 scope global lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host proto kernel_lo
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1300 qdisc mq state UP group default q
len 1000
    link/ether 00:15:5d:55:49:0b brd ff:ff:ff:ff:ff:ff
    altname enx00155d55490b
    inet 172.20.206.173/20 brd 172.20.207.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::215:5dff:fe55:490b/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
praveen@praveenasus:~$
```

Step 3: Connect to the Linux VM Using SSH

From another Linux machine or the Windows host (using an SSH client):

•ssh praveen@172.20.206.173

```
praveen@praveenasus:~$ ssh praveen@172.20.206.173
The authenticity of host '172.20.206.173 (172.20.206.173)' can't be established.
ED25519 key fingerprint is: SHA256:K8CM02KRJwQCULbFm++7MNHaGH/r5aP/Tik8EkoD+sA
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: y
Please type 'yes', 'no' or the fingerprint: n
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '172.20.206.173' (ED25519) to the list of known hosts.
praveen@172.20.206.173's password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 6.6.114.1-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Sat Jun 20 06:05:16 UTC 2026

  System load:  0.0                 Processes:             42
  Usage of /:   0.2% of 1006.85GB   Users logged in:       0
  Memory usage: 14%                 IPv4 address for eth0: 172.20.206.173
  Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge
Welcome to Ubuntu 26.04 LTS (GNU/Linux 6.6.114.1-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:     https://landscape.canonical.com
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 System information as of Sat Jun 20 06:05:16 UTC 2026

  System load:  0.0                 Processes:             42
  Usage of /:   0.2% of 1006.85GB   Users logged in:       0
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  Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

This message is shown once a day. To disable it please create the
/home/praveen/.hushlogin file.
praveen@praveenasus:~$
```

Step 4: Generate an SSH Key Pair

Generate a public/private key pair:

•ssh-keygen

Press **Enter** to accept the default location.

Optionally set a passphrase, or press **Enter** to leave it empty.

```
praveen@praveenasus:~$ ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/praveen/.ssh/id_ed25519):
Enter passphrase for "/home/praveen/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/praveen/.ssh/id_ed25519
Your public key has been saved in /home/praveen/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:pg2NQ5rVvjxNKJvjaaykonkYK6lfGnJUhGYH1yVHeFc praveen@praveenasus
The key's randomart image is:
+--[ED25519 256]--+
|   .oo..++   .E  |
|   +o. o+. .     |
| o .. o...       |
|    . = + .      |
|   . o = S .     |
|..     @ +       |
|.=o o.= = .      |
|*+.* .oo .       |
|B++ .oo          |
+----[SHA256]-----+
praveen@praveenasus:~$
```

Step 5: Copy the Public Key to the Remote Server

Use:

•ssh-copy-id praveen@172.20.206.173

This command copies the contents of **id_rsa.pub** to the remote user's ~/.ssh/authorized_keys file.

```
praveen@praveenasus:~$ ssh-copy-id praveen@172.20.206.173
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/praveen/.ssh/id_ed25519.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already
 installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install
the new keys
praveen@172.20.206.173's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'praveen@172.20.206.173'"
and check to make sure that only the key(s) you wanted were added.

praveen@praveenasus:~$
```

Step 6: Login Without a Password

Now connect again:

•ssh praveen@172.20.206.173

```
praveen@praveenasus:~$ ssh praveen@172.20.206.173
Welcome to Ubuntu 26.04 LTS (GNU/Linux 6.6.114.1-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Sat Jun 20 06:08:04 UTC 2026

  System load:  0.0                 Processes:             49
  Usage of /:   0.2% of 1006.85GB   Users logged in:       0
  Memory usage: 15%                 IPv4 address for eth0: 172.20.206.173
  Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge
Last login: Sat Jun 20 06:05:17 2026 from 172.20.206.173
praveen@praveenasus:~$
```

If the key is configured correctly, SSH authenticates using the private key and logs you in without asking for the account password (unless you protected the key with a passphrase).

Summary

Command	Purpose
sudo apt install openssh-server	Installs the SSH server
sudo systemctl start ssh	Starts the SSH service
sudo systemctl enable ssh	Starts SSH automatically at boot
sudo systemctl status ssh	Checks the SSH service status
ip addr	Displays the system's IP address
ssh user@ip	Connects to a remote Linux machine
ssh-keygen	Generates an SSH key pair
ssh-copy-id user@ip	Copies the public key to the remote server

Command	Purpose
ssh user@ip	Logs in using key-based authentication

Task – 5: SSH Hardening & RDP Overview

SOLUTION-

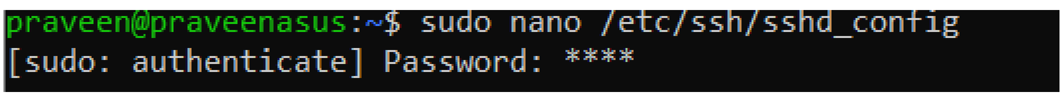
➤SSH Hardening

Introduction

SSH (Secure Shell) is the most commonly used protocol for remote Linux administration. Since SSH provides remote access to a system, it is often targeted by attackers. SSH hardening improves security by restricting access and reducing common attack vectors.

Step 1: Open the SSH configuration file

sudo nano /etc/ssh/sshd_config



Step 2: Make only these three changes

PermitRootLogin no
Port 2222
AllowUsers praveen

(Replace praveen with your username if it's different.)

```
  GNU nano 8.7.1                              /etc/ssh/sshd_config *
# default value.

# In general, the first uncommented definition of an option takes precedence.
# For options that accept multiple values, like 'Port', subsequent definitions
# are appended to the configuration.
#
# Note that the above implies that configuration options from snippets in
# /etc/ssh/sshd_config.d/*.conf take precedence over those defined in this
# file. In addition, configuration snippet files are processed in lexical
# order, so options defined in files with names that sort earlier take
# precedence.
#
# Such configuration snippets may be present in default installations of
# Ubuntu.
Include /etc/ssh/sshd_config.d/*.conf

# When systemd socket activation is used (the default), the socket
# configuration must be re-generated after changing Port, AddressFamily, or
# ListenAddress.
#
# For changes to take effect, run:
#
#   systemctl daemon-reload
#   systemctl restart ssh.socket
#
#Port 2222
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
#PermitRootLogin no
```

```
# PAM authentication via KbdInteractiveAuthentication may bypass
# the setting of "PermitRootLogin prohibit-password".
# If you just want the PAM account and session checks to run without
# PAM authentication, then enable this but set PasswordAuthentication
# and KbdInteractiveAuthentication to 'no'.
UsePAM yes

#AllowAgentForwarding yes
#AllowTcpForwarding yes
#GatewayPorts no
X11Forwarding yes
#X11DisplayOffset 10
#X11UseLocalhost yes
#PermitTTY yes
PrintMotd no
#PrintLastLog yes
#TCPKeepAlive yes
#PermitUserEnvironment no
#Compression delayed
#ClientAliveInterval 0
#ClientAliveCountMax 3
#UseDNS no
#PidFile /run/sshd.pid
#MaxStartups 10:30:100
#PermitTunnel no
#ChrootDirectory none
#VersionAddendum none

# no default banner path
#Banner none

# Allow client to pass locale and color environment variables
AcceptEnv LANG LC_* COLORTERM NO_COLOR

# override default of no subsystems
Subsystem       sftp    /usr/lib/openssh/sftp-server

# Example of overriding settings on a per-user basis
#Match User anoncvs
#       X11Forwarding no
#       AllowTcpForwarding no
#       PermitTTY no
#       ForceCommand cvs server
AllowUser praveen
```

Step 3: Save the file

- Press **Ctrl + O**
- Press **Enter**
- Press **Ctrl + X**

Step 4: Check for errors

•sudo sshd -t

If there is **no output**, the configuration is correct.

```
praveen@praveenasus:~$ sudo sshd -t
praveen@praveenasus:~$
```

Step 5: Restart SSH

sudo systemctl restart ssh

```
praveen@praveenasus:~$ sudo systemctl restart ssh
praveen@praveenasus:~$
```

Step 6: Verify

sudo systemctl status ssh

```
praveen@praveenasus:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
     Active: active (running) since Sat 2026-06-20 06:27:53 UTC; 29s ago
 Invocation: 0d6b20d9c39149648101dc0aff8f6d5f
TriggeredBy: ● ssh.socket
       Docs: man:sshd(8)
             man:sshd_config(5)
    Process: 7528 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 7531 (sshd)
      Tasks: 1 (limit: 4333)
     Memory: 1.2M (peak: 2.3M)
        CPU: 45ms
     CGroup: /system.slice/ssh.service
             └─7531 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jun 20 06:27:53 praveenasus systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Jun 20 06:27:53 praveenasus sshd[7531]: Server listening on 0.0.0.0 port 22.
Jun 20 06:27:53 praveenasus sshd[7531]: Server listening on :: port 22.
Jun 20 06:27:53 praveenasus systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
praveen@praveenasus:~$
```

Step 7: Connect using the new port

ssh -p 2222 praveen@<your-ip-address>

```
praveen@praveenasus:~$ ssh -p 2222 praveen@172.20.206.173
```

Short Theory (for your notes)

SSH Hardening is the process of improving the security of the SSH service. It protects the system from unauthorized access and common attacks. In this task:

- PermitRootLogin no disables direct root login.
- Port 2222 changes the default SSH port from 22 to 2222.
- AllowUsers praveen allows only the specified user to log in through SSH.

➤ RDP (Remote Desktop Protocol)

Introduction

RDP (Remote Desktop Protocol) is a network protocol developed by Microsoft that allows users to remotely access and control another computer's graphical desktop environment.

Unlike SSH, which provides command-line access, RDP provides a complete graphical desktop interface.

RDP is commonly used in Windows environments but can also be used on Linux through software such as **xrdp**

Installing xrdp on Ubuntu

Step 1: Update Repository

sudo apt update

```
praveen@praveenasus:~$ sudo apt update
Hit:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu resolute InRelease
Hit:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
14 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

Step 2: Install xrdp

sudo apt install xrdp -y

```
praveen@praveenasus:~$ sudo apt install xrdp -y
Installing:
  xrdp

Installing dependencies:
  cpp                         libunwind8        libxxf86dga1        xorgxrdp
  cpp-15                      libutempter0      luit                xserver-common
  cpp-15-x86-64-linux-gnu     libxaw7           x11-apps            xserver-xorg
  cpp-x86-64-linux-gnu        libxcb-damage0    x11-session-utils   xserver-xorg-core
  gcc-15-base                 libxcb-dri2-0     x11-utils           xserver-xorg-legacy
  libdrm-nouveau2             libxcb-shape0     x11-xkb-utils       xserver-xorg-video-all
  libdrm-radeon1              libxcb-util1      x11-xserver-utils   xserver-xorg-video-amdgpu
  libfontenc1                 libxcvt0          xbitmaps            xserver-xorg-video-ati
  libglu1-mesa                libxfont2         xcvt                xserver-xorg-video-fbdev
  libice6                     libxft2           xfonts-base         xserver-xorg-video-intel
  libisl23                    libxkbfile1       xfonts-encodings    xserver-xorg-video-nouveau
  libjpeg-turbo8              libxmu6           xfonts-scalable     xserver-xorg-video-qxl
  libjpeg8                    libxpm4           xfonts-utils        xserver-xorg-video-radeon
  libmpc3                     libxss1           xinit               xserver-xorg-video-vesa
  libopengl0                  libxt6t64         xinput              xterm
  libopus0                    libxv1            xorg
  libsm6                      libxvmc1          xorg-docs-core

Suggested packages:
  cpp-doc          mesa-utils   xfonts-100dpi  | xfonts-75dpi            firmware-misc-nonfree
  gcc-15-locales   nickle       xfonts-75dpi   firmware-amd-graphics     xfonts-cyrillic
  cpp-15-doc       cairo-5c     x11-xfs-utils  xserver-xorg-video-r128
  opus-tools       xorg-docs    xfonts-100dpi  xserver-xorg-video-mach64

Summary:
  Upgrading: 0, Installing: 67, Removing: 0, Not Upgrading: 14
  Download size: 31.5 MB
  Space needed: 80.6 MB / 1024 GB available

Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 libjpeg-turbo8 amd64 2.1.5-4ubuntu4 [153 kB
Get:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 libjpeg8 amd64 8c-2ubuntu12 [2142 B]
Get:3 http://archive.ubuntu.com/ubuntu resolute/main amd64 libopus0 amd64 1.6.1-1 [3564 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute/universe amd64 xrdp amd64 0.10.1-4.1 [669 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute/main amd64 gcc-15-base amd64 15.2.0-16ubuntu1 [39.0 kB
```

Step 3: Start xrdp Service

sudo systemctl start xrdp

Step 4: Enable xrdp at Boot

sudo systemctl enable xrdp

Step 5: Verify Service

sudo systemctl status xrdp

```
praveen@praveenasus:~$ sudo systemctl start xrdp
praveen@praveenasus:~$ sudo systemctl enable xrdp
Synchronizing state of xrdp.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable xrdp
praveen@praveenasus:~$ sudo systemctl status xrdp
● xrdp.service - xrdp daemon
     Loaded: loaded (/usr/lib/systemd/system/xrdp.service; enabled; preset: enabled)
     Active: active (running) since Sat 2026-06-20 06:44:43 UTC; 52s ago
 Invocation: 0246ae0984fb4c7e9db9e439c8994f9a
       Docs: man:xrdp(8)
             man:xrdp.ini(5)
   Main PID: 8956 (xrdp)
      Tasks: 1 (limit: 4333)
     Memory: 1M (peak: 2.2M)
        CPU: 21ms
     CGroup: /system.slice/xrdp.service
             └─8956 /usr/sbin/xrdp --nodaemon

Jun 20 06:44:43 praveenasus systemd[1]: Starting xrdp.service - xrdp daemon...
Jun 20 06:44:43 praveenasus systemd[1]: Started xrdp.service - xrdp daemon.
Jun 20 06:44:43 praveenasus xrdp[8956]: [INFO ] starting xrdp with pid 8956
Jun 20 06:44:43 praveenasus xrdp[8956]: [INFO ] address [0.0.0.0] port [3389] mode 1
Jun 20 06:44:43 praveenasus xrdp[8956]: [INFO ] listening to port 3389 on 0.0.0.0
Jun 20 06:44:43 praveenasus xrdp[8956]: [INFO ] xrdp_listen_pp done
```

Step 6: Allow RDP Through Firewall

sudo ufw allow 3389/tcp

```
praveen@praveenasus:~$ sudo ufw allow 3389/tcp
Rules updated
Rules updated (v6)
```

Step 7: Find System IP Address

ip addr

```
praveen@praveenasus:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet 10.255.255.254/32 brd 10.255.255.254 scope global lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host proto kernel_lo
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1300 qdisc mq state UP group default qlen 1000
    link/ether 00:15:5d:55:49:0b brd ff:ff:ff:ff:ff:ff
    altname enx00155d55490b
    inet 172.20.206.173/20 brd 172.20.207.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::215:5dff:fe55:490b/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
```

➥Connecting Using RDP

From Windows

1. Press **Windows + R**
2. Type:

mstsc

3. Click OK
4. Enter Linux IP Address

172.20.206.173

5. Click Connect
6. Enter Linux username and password

Example:

Username: praveen
Password: ********

The Ubuntu desktop appears remotely.